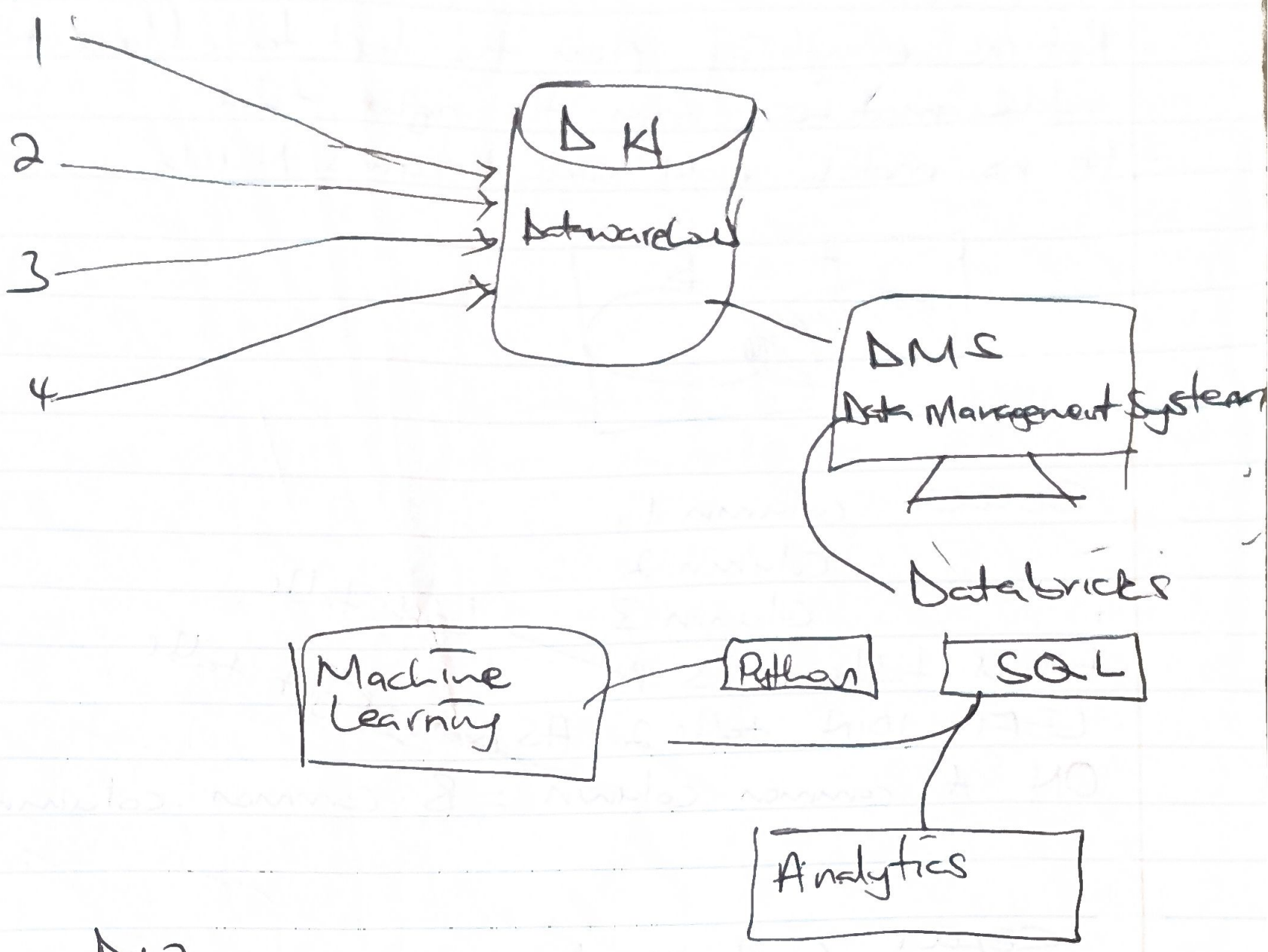
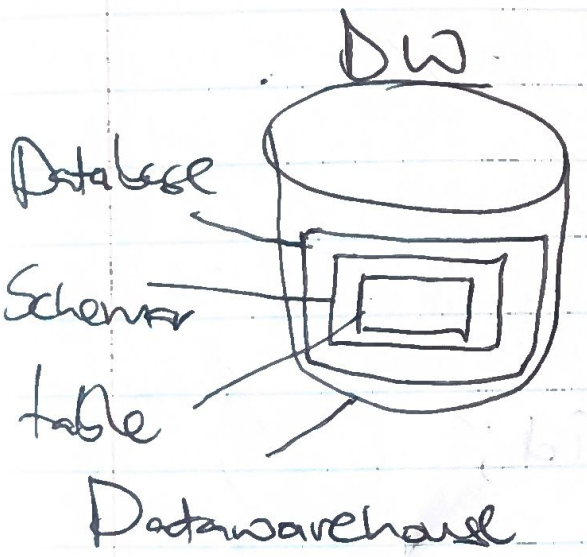
SQL Standard Part 1

Data →

Different Data Sources

ETL (extract transform load)





+ Inside the Datawarehouse you have multiples database.
* Data warehouse is our central data store

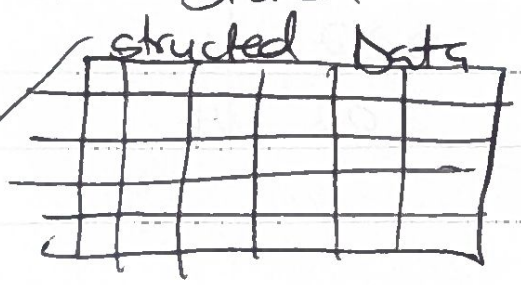
Databricks is my data management system

Catalog is the data warehouse
Workspace is the database
Default is the schema
then table

(sample/backhaul)

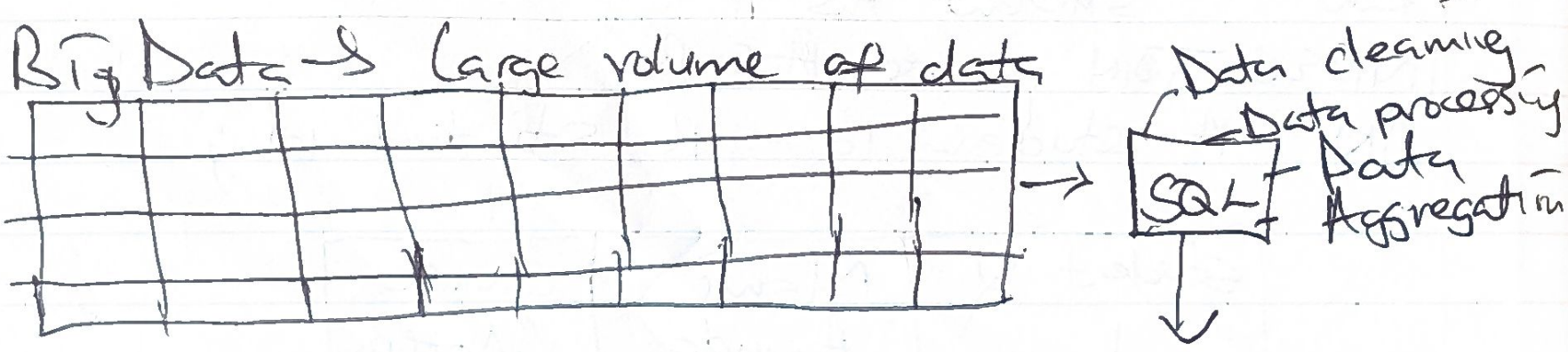
In one data warehouse you can have multiple databases
In one database you can have multiple schemas
In on schema you can have multiple tables

Data is stored in table (rows & columns)

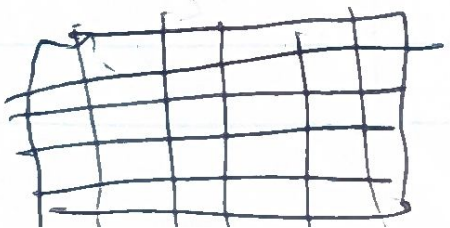


Structed Structured Query Language (SQL)
We use it to extract insights from our data

- We use SQL for structured data -> it is data in a table.
- Insights from our is key information that help us to make decision based on data.



Aggregate. data
Summarizing data



SQL JOIN

- In order to join two or more tables you need to have a common column
- Joining table give us the ability to select any column between the tables that have been joined together.
- The tables joined because one table
- Joins affect the rows - columns not affected.

INNER JOIN

- Returns the rows that are matching in both tables

Students

Students id	Name
1	Ayanda
2	Kabelo
3	Zanele

Course

Course Id	Student id	Cours
201	1	Ma
202	1	Scie
203	4	Engl

```
SELECT columns (s)
FROM   table1 AS A
INNER JOIN table2 AS B
ON A.common_column = B.Common column

SELECT    Student id,
          Name,
          Course
FROM      Student AS A
INNER JOIN course AS B
ON   A.student id = B.student id;
```

Student Id	Name	Course
1	Ayanda	Mate
3	Zanele	Scien

Delete

Drop and truncate are danger (runaway).

LEFT JOIN

want every row in the left table (base table)
want only the rows that match from the other table.

```
SELECT column
From table1 AS A
LEFT JOIN table 2 AS B
ON A.common_column = B common_column;
```

Sales Customer

Customer_id	Name
1	Rofhiwa
2	Mvuko

orders

order id	custome id	Amount
101	1	300
102	2	150
103	3	500

Question

Write an SQL script to get the customer_id, name, amount.

```
SELECT customer id,
       Name,
       Amount
FROM Customers Ac A.
LEFT JOIN orders AS B
ON A. customer id = B. customer id
```

customer	Name	Amount
1	Rofhiwa	300
2	Mvuko	150

Departments

dept_id	dept_name
10	Sales
20	IT
30	HR
40	Legal

Employees

emp_id	emp_name	dept_id
1	Neo	10
2	Rofhi	20
3	Thabo	20
4	Lesedi	50

I want the dept_id, dept_name, emp_name.

```
SELECT emp_id
       dept_id,
       Dept_nam,
       Emp_nanae.
FROM employees AS A
LEFT JOIN dept AS B
ON A dept id = B dept_id;
```

Emp id	Emp name	dept_id	dept name
1	Neo	10	Sales
2	Rofhi	20	IT
3	Thabo	20	H
4	Lesedi	50	NULL

Students

student id	Name
1	Sam
2	Lulu
3	Ben

club

club_id	student id	club name
10	1	Chess
11	2	Debate
12	4	Soccer

Question

I want to know all the students that play sports, and what sport they play

```
SELECT   Student id,
         club name,
         Name
FROM     Students AS A
LEFT JOIN Clubs AS B
ON       Student id = B student id
```

club_id	student_id	Name	club name
0	1	Sam	Chess
1	2	Lulu	Debate
2	3	Bird	Soccer
			NULL

NULL → empty space
& unknown value
& Missing value
→ Blank

FULL OUTER JOIN
- Returns all the rows from both the tables
- Because it is a join you can select any column from the tables you are

A FULL OUTER JOIN returns everything from both tables
- If a row matches (same key on both tables (side)
- It combines them into one row
- If a row exists only in the left table →
  It still show up, with NULLS for the right table columns

Syntax:

```
SELECT column
FROM table1 AS A
FULL OUTER JOIN table 2 AS B
ON A. common_column = B common-column;
```

Student

Students_id	Name
1	Sam
2	Lulu
3	Ben

Sports

Sports_id	student_id	sport_name
10	1	Chess
11	2	Debate
12	4	Soccer

```
SELECT student_id,
       name,
       Sport_id,
       Sport name
FROM student AS A
FULL OUTER JOIN sports AS B
ON Astudent_id = B. student_id;
```

student_id	Name	sports_id	sport_name
1	Sam	10	Chess
2	Lulu	11	Debate
3	Ben	NULL	Nul
4	NULL	12	Soccer

Department

dept_id	dept_name
10	Sales
20	IT
30	HR
40	Legal

Employees

Empl_id	Emp_name	Dept_id
1	Noo	10
2	Amina	20
3	Thabo	20
4	Mia	50

- Write a SQL script to give me all the employee names and all the department names in the company.

Dept id

INNER JOIN

~~Project~~	Employee id	First name	Fist	Last	Dep	Salary	Project

Sedect 1 employees, A first name, DepartA.
FROM Table 1 AS A
INNER JOIN Table 2 AS B
ON A employees_id = B employees_id.

Employee	First	Last	Department	Salary	Project	Project Name	Budget	Status
1	John	Doe	IT	70000	101	It Devel	100 000	Complete
2	Alice	Smith	HR	60 000	102	Employ	50000	Ongoing
3	Bob	John	Finance	75 000	104	Financial	90 000	Ongoing
5	Emma	Wilson	Sales	65000	105	Market	65 000	Complete
6	Michael	Clark	Finance	80 000	106	Risk M	80 000	Pending

Select employees, First name, Departme.
FROM Table 1 AS A
LEFT JOIN Table 2 AS B.
ON A. employees_id = B employees_id

Case statement

-> Help us to create flags in our data
-> Help us to build logical conditions
-> A case statement is normally created within a SELECT statement
-> A case statement creates a new column.

```
SELECT
        CASE WHEN condition 1 THEN result 1
        WHEN Condition 2 THEN result 2
    ELSE default_result
    END AS new_ Column_name
FROM table;
```

You will get a default_result when all the outlined conditions are not satisfied

Sales

sale_id	store	Price Amount
1	Pep	100
2	Pep	50
3	Mr Price	80
4	Mr Price	20

0 - 40 low price
41 - 60 Med price
> 60 High price

```
Select  sale_id ; store , price
    Case
        When price BETWEEN 0 AND 40 THEN 'Low Price'
        When price BETWEEN 41 AND 60 THEN 'Med Price'
        ELSE 'High Price'
        END AS price_group
```

Sale_id	Store	Price	Price_group
1	Pep	100	High price
2	Pep	50	med
3	Mr Price	80	High
4	Mr Price	20	Low

```
Select Product_name, price, product_Id,
    Case
    When Expensive then Price > 1000.
    When Mid_range then Price between 100 and 1000
    ELSE Budget Price < 100.
    END AS
```

Product_Id	Product_name	Price	Price category
1	Laptop	1200	Expensive
2	Phone	800	Mid
3	Keyboard	45	Budget
4	Monitor	300	Mid
5	Mouse	25	Budget

A case statement helps us to classify, to group, to bucket our data
A case statement help us to create a categorical column from a numeric column based on specific conditions.

```
SELECT product_id, product_name, price,
 Case  when price > 1000 THEN 'expensive'
       When price Between 100 and 1000 THEN 'Mid
       WHEN price < 100 THEN 'Budget'
 END AS price_category
 FROM products;
```

Lesson 20

TP iMiro is the mind map.

The coffee shop has not been generating sales. Therefore the business appointed a new CEO to help growing the sales and revenue (qty * price).

My duty as a Data Analytics is to extract actionable insights from historical data and prepare a presentation to assist the CEO in decision-making.

The insights will give us the recommendation as to what can be done in order to increase the revenue and improve product performance.

Tools allowed

You may use any tool at your disposal e.g project planning Miro, figma

coding Platforms

Microsoft SQL Server ,

Databricks

Presentation & Reporting

Canva

Miro

Data Visualization

Microsoft Excel , Power BI

Task 1

Inspect the data

① Categorical data
Store Location , Product category , Product Detail
Product Det

② Quantitative Data
Transaction Qty , Unit Price (Revenue = Quantity x Price)

LESSON 21 (Miro)
What is it that we I-m gonna do with ~~the data that~~ I have?
- Plan the flow chart, the flow of the data
NOTE: We need to understand where is the data comming from and where is it going.
- Building the concept map
- Map out the diagram of the flow of data (where is the data sitting & where should it go and how has been used in each and every step

~~Where is the data Source~~

Analysing the process flow on Miro

CANVA.com
It outlines when we gonna do what
After what we gonna do (Miro) then when we gonna do (canva)

Advanced SQL

Null functions

What is a NULL?

- No value
- Missing value
- No data
- Empty space
- Blank
- No record
- Empty
- Missing information

Null are functions that helps us to deal with null value in our dataset.
Null fuctions help us to handle null values

CASE1 : How to check for NULL values in a specific table on a specific column

```
IS NULL
Select columns
From table_name
WHERE column_name IS NULL;
```

↳ This code is to check if you have a NULL value in a specific column

Fruit_sales

Sales id	Product	Amount
1	Banana	100
2	Apple	NULL
3	Orange	20
4	Grapes	40
5	lemons	NULL

```
SELECT product, amount
FROM   fruits_sales
WHERE  amount IS NULL;
```

Product	Amount
Apple	NULL
Lemons	NULL

IS NOT NULL

CASE 2: We want to find the rows where a special column do not have null values:

```
Select *
FROM table_name
WHERE column_name IS NOT NULL;
```

```
Select product, amount
FROM   fruits_sales
WHERE  amount IS NOT NULL;
```

Product	Amount
Banana	100
Orange	20
Grapes	40

CASE 3: Replacing a NULL value ~~in a specific~~ with a default value

IF NULL (column_name, replacement)

→ Placed within a select statement used with AS to give it a descriptive name.

```
SELECT sale_id, product,
       IFNULL(amount, 0) As Amount
FROM fruit_sales ;
```

Sale id	Product	Amount
1	Banana	100
2	Apple	0
3	Orange	20
4	Grape	40
5	Lemons	0

Ass 4: We want to have multiple alternative value

COALESCE (exp1, exp2, exp3, exp4 ----) eg: Column 1, 2,]

= Returns the first non-null value from a list of expressions
- It checks the expression from left-right

Boxing team

Name	Student_id	ID_no
Rafhi	101	NULL
Mark	NULL	111
Elon	103	112
Jeff	102	NULL
Larry	NULL	114
Klarren	NULL	NULL

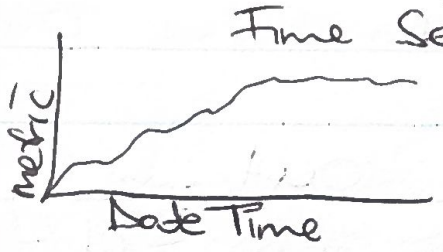
Outcome

Name	Unique_id
Rafhi	101
Mark	111
Elon	103
Jeff	102
Larry	114
Klarren	0

```
Select name,
    COALESCE(student id; id_no ; 0)
From boxing_team
```

Google & Big Query & try it

1. **Date functions** → It helps us to deal with the date
Indicates a specific period
Indicates an event
Data functions are functions that help us to handle dates in our data
What can we do with data functions
We can extract the
year, day, week, month, hour, minutes, Quarters, Seconds
A date itself is a data type
Date functions are found within a select statement (and create a new column)
OLTP database
Captures transactions at a specific period of time



We need to understand the Date functions so that we can handle or manipulate the date in our data

① Year → It is used to extract the year on the date column
Select Year('2025-06-02') As Year_of_love

Year_value
2025

- We normally use the date functions within the select statement
- A function placed within the select statement creates a new column
- We have to use the As to rename the column whenever we apply a function on it.

② Month() → Extract the month.
Select month ('2025-06-02') As month_value;

month_value
6

③ Day() – Extracts the day
Select Day('2025-06-02') As Day_Value;

④ DayName() Select Dayname('2025-06-02') As Dt

Dt
Monday

Extract mon_sun based on the System Calendar (Time zone)

⑤ Monthname() Select Monthname('2025-06-02') As M
Extracts month name as in Jan-Dec based on the syste
time zone (DMS) eg databricks

Jun

DateADD(unit, value, date) → represents increment or decreament.
↑ Date column.
It is used to add or subtract a date interval
Unit – Day
– Month
– Year
Select DateADD(Day, 2, '2025-06-02') As 2_later

2_later
2025-06-04

③ Month() - Extract the month
Select month ('2025-06-02') As month_value;

month_value
6

③ Day() - Extracts the day
Select Day ('2025-06-02') As Day_Value;

④ DayName() Select Dayname ('2025-06-02') As Dt

Dt
Monday

Extract noon_sun based on the system Calendar (Time zone)

⑤ Monthname() Select Monthname ('2025-06-02') As M
Extracts month name as in Jan-Dec based on the syste
time zone (DMS) eg databricks

Jun

DateADD (unit, value, date) → represents increment or decreament
Date column.
It is used to add or subtract a date interval
Unit - Day
- Month
- Year
Select DateADD (Day, 2, '2025-06-02') As 2_later

2_later
2025-06-04

* DateDiff (unit, date1, date2)
Used to calculate the difference between two dates
date 2 - date 1 = value
Unit - Day, month, Year
Select DateDiff (Year, '2025-06-02', 2027-06-02)
2

* Current_date() → It gives you today's date based on the DMS settings

Select Current_Date();
'2026-04-13'

Current_Timestamp();
2026-04-13 20:05:54:13
will give you the date and the time

What you need on your readme file:

1. Problem statement
2. Aim of the project
3. Objective - steps taken to solve for the aim.
4. Summary of results
5. Tools used in the case study.